

SAM3 prompting ablation -- mask AP per class, test split (source: recomputed)

class	SAM3 text AP@0.5	SAM3 text AP@0.5:0.95	SAM3+YOLO AP@0.5	SAM3+YOLO AP@0.5:0.95	SAM3 GTbox AP@0.5	SAM3 GTbox AP@0.5:0.95	YOLO11n AP@0.5	YOLO11n AP@0.5:0.95
base of tongue	0.105	0.060	0.424	0.280	0.974	0.701	0.424	0.288
bipolar	0.025	0.018	0.505	0.505	1.000	0.742	0.505	0.505
bot	0.131	0.105	1.000	0.806	1.000	0.810	1.000	0.903
cut	0.007	0.003	0.616	0.116	1.000	0.428	0.779	0.222
endotracheal tube	0.000	0.000	1.000	0.903	1.000	0.902	1.000	0.898
ligasure	0.000	0.000	0.000	0.000	1.000	0.724	0.000	0.000
maryland	0.062	0.038	0.434	0.294	0.951	0.692	0.440	0.315
medial pterygoid muscle	0.000	0.000	0.692	0.418	1.000	0.575	0.692	0.336
monopolar	0.000	0.000	0.357	0.306	0.976	0.696	0.357	0.324
parapharyngeal fat	0.000	0.000	0.779	0.453	1.000	0.549	0.812	0.512
posterior pharyngeal wall	0.000	0.000	0.435	0.196	0.978	0.673	0.468	0.176
prevertebral fascia	0.000	0.000	0.000	0.000	1.000	0.700	0.000	0.000
soft palate	0.000	0.000	0.184	0.130	0.735	0.474	0.556	0.397
suction	0.069	0.054	0.559	0.426	0.961	0.695	0.568	0.427
mean (aggregate)	0.028	0.020	0.499	0.345	0.970	0.669	0.543	0.379

SAM3 text: class-name text prompts: detect + classify + segment, zero-shot

SAM3+YOLO: YOLO11n boxes (+class, +score) as prompts: SAM3 only draws the mask

SAM3 GTbox: GT boxes as prompts: perfect localisation + class = upper bound

YOLO11n: reference: the detector whose boxes feed the row above